

BATTERY BACKUP CALCULATION REPORT

Security Power Supplies (LSP Enclosures) · Division 28 · Rev 3

St. Ignatius College Preparatory · 2001 37th Ave, San Francisco, CA

General Contractor: Devcon · Consultant: TEECOM · Spec: Section 28 00 00

Parameter	Value / Basis
Prepared by	Dynamic Security Technologies
Scope	Standby backup for the 8 LSP FlexPower enclosures (one per MDF/IDF closet). Covers ACAMS field power (24 VDC) and the intrusion field devices that share the LSP 12 VDC supply.
Excluded	Bosch B9512G intrusion panel — supplied by its own dedicated AC transformer and battery backup (separate calculation).
Backup Requirement	≥ 4-hour standby backup on batteries alone (upon AC power loss)
Report Date	July 20, 2026
Revision	Rev 3 — adds intrusion field-device load on the 12 VDC bus (per Intrusion Loading Schedule V4)

1. Purpose & Backup Condition

This report documents secondary (battery) backup calculations for the eight LSP FlexPower enclosures at St. Ignatius, per Section 28 00 00 (Addendum 6, TEECOM, May 14, 2025), requiring ≥ 4 hours standby backup for all security equipment and locks.

Backup condition modeled — Loss of AC power: on utility power failure, each LSP supply AC input and battery charger are offline and the entire connected load runs on the sealed batteries alone. The 24 VDC loads are carried by two series-connected 12 V batteries (24 V @ 12 Ah); the 12 VDC loads by the single 12 V battery (12 Ah). No reliance on AC during the 4-hour period.

2. Battery Configuration (per Enclosure)

Power Supply	Bus	Battery Configuration	Capacity	Serves
FPO-250 (250 W)	24 VDC	(2) 12V 12Ah in SERIES	12.0 Ah	ACAMS card readers + fail-safe maglocks
FPO-150 (150 W)	12 VDC	(1) 12V 12Ah	12.0 Ah	Intrusion field devices + SDI2 boards

3. Methodology & Assumptions

Backup (hours) = Usable Capacity (Ah) ÷ Connected Load (A). Usable = 12.0 Ah × 0.80 derating = 9.6 Ah. 4-hour load budget per bus = 9.6 ÷ 4 = 2.40 A.

- Nominal (standby) current is used for all devices.
- ACAMS: card readers (Schlage MTB) = 45 mA @ 24V; fail-safe maglocks = 250 mA each (energized-to-lock, drawn continuously); fail-secure locks = 0 A; wireless locks = 0 A; door-release buttons = 0 A; DH excluded. Reader counts per closet from Drawing T0.05.
- Maglock allowance = 22 (3 per Lower/L1/L3 closet, 2 per L2 closet), on the 24 VDC bus.
- Intrusion field devices on the 12 VDC bus (per Intrusion Loading Schedule V4): keypad B942 = 200 mA standby (proximity reader disabled — add 100 mA/keypad if enabled); motion CDL2-15G = 15 mA; glass-break DS1101i = 23 mA; duress button = 0 A (mechanical dry contact).
- SDI2 boards on the 12 VDC bus: B208 octo-input ("OctoPOPIT") = 35 mA each; B299 POPEX = 35 mA each; D9127 single POPIT = 0.8 mA each.

- Points labeled "FROM ACP" (LA door-alarm, DC door contacts, CR/WR fault outputs) are dry-contact signals into the Bosch panel — 0 A on the 12 VDC bus. The Bosch B9512G panel and its own battery are excluded.

4. 24 VDC Bus (FPO-250) — ACAMS: Readers + Fail-Safe Maglocks

Enclosure	Readers	Maglocks	Load (A)	Backup	Result
MDF-0.1	6	3	1.020	9.4 h	✓ PASS
IDF-0.2	10	3	1.200	8.0 h	✓ PASS
IDF-1.1	15	3	1.425	6.7 h	✓ PASS
IDF-1.2	10	3	1.200	8.0 h	✓ PASS
IDF-2.1	5	2	0.725	13.2 h	✓ PASS
IDF-2.2	7	2	0.815	11.8 h	✓ PASS
IDF-3.1	9	3	1.155	8.3 h	✓ PASS
IDF-3.2	11	3	1.245	7.7 h	✓ PASS

✓ All 8 enclosures PASS on batteries alone. Worst: IDF-1.1 = 1.425 A → 6.7 h (uses 59% of the 2.40 A budget). Totals: 73 readers + 22 maglocks.

5. 12 VDC Bus (FPO-150) — Intrusion Field Devices + SDI2 Boards

Enclosure	KP	MD	GB	DB	B208	B299	POPIT	Load A	Bkp h	OK
MDF-0.1	0	0	4	0	1	0	0	0.127	75.6	✓
IDF-0.2	2	2	2	0	3	0	0	0.581	16.5	✓
IDF-1.1	1	1	16	8	4	1	16	0.771	12.5	✓
IDF-1.2	3	1	15	0	2	1	15	1.077	8.9	✓
IDF-2.1	0	0	1	0	1	0	0	0.058	165.5	✓
IDF-2.2	0	0	0	0	1	0	0	0.035	274.3	✓
IDF-3.1	0	0	9	0	2	1	9	0.319	30.1	✓
IDF-3.2	0	0	6	0	1	1	6	0.213	45.1	✓

✓ All 8 enclosures PASS. Worst: IDF-1.2 = 1.077 A → 8.9 h (45% of the 2.40 A budget). Totals on the 12 VDC bus: 6 keypads, 4 motions, 53 glass-breaks, 8 duress (0 A), 15 B208, 4 B299, 46 POPITs. Keypad draw dominates this bus; even at 300 mA/keypad (prox enabled) the worst enclosure (IDF-1.2) rises to ~1.38 A → ~7.0 h, still within budget.

6. Result Summary

Every LSP enclosure meets the 4-hour standby requirement on both buses, on batteries alone, including the 22-maglock allowance and the full intrusion field-device load. Governing (shortest) backup at each enclosure:

Enclosure	24V backup	12V backup	Governing	Result
MDF-0.1	9.4 h	75.6 h	9.4 h	✓ PASS
IDF-0.2	8.0 h	16.5 h	8.0 h	✓ PASS
IDF-1.1	6.7 h	12.5 h	6.7 h	✓ PASS
IDF-1.2	8.0 h	8.9 h	8.0 h	✓ PASS
IDF-2.1	13.2 h	165.5 h	13.2 h	✓ PASS
IDF-2.2	11.8 h	274.3 h	11.8 h	✓ PASS
IDF-3.1	8.3 h	30.1 h	8.3 h	✓ PASS
IDF-3.2	7.7 h	45.1 h	7.7 h	✓ PASS

Governing worst case across the project: 6.7 hours (IDF-1.1, 24 VDC bus) — 2.7 hours above the 4-hour requirement.

7. Notes

- Fail-safe maglocks are energized while locked and draw current continuously; fail-secure locks draw 0 A in standby. Only fail-safe devices consume battery during backup.
- Keypad standby current assumes the B942 built-in proximity reader is disabled (200 mA). If enabled for arm/disarm, use 300 mA — all enclosures still pass (see Section 5).

- Reader quantities per closet are from Drawing T0.05; intrusion device and SDI2-board quantities and closet assignments are from the Intrusion Loading Schedule V4.
- A 0.80 usable-capacity derating is applied (VRLA aging/temperature); at beginning-of-life all durations are ~25%% longer.
- The Bosch B9512G panel is on its own AC transformer and battery and is not part of this calculation. Local audible alarms (ES4200), door contacts, and card-reader fault outputs appear on the Bosch panel only as dry-contact input points (0 A on the LSP 12 VDC bus).
- ACAMS head-end controllers (HID Mercury MP1502/MR52), if powered from an LSP 12 VDC bus, are to be confirmed against the panel power schedule; spare 12 VDC capacity remains in every enclosure to accommodate them.

8. Certification & Sign-Off

The undersigned certifies that each LSP enclosure provides not less than four (4) hours of standby backup on batteries alone (loss-of-AC), on both the 24 VDC and 12 VDC buses, for the connected ACAMS load (readers + 22 fail-safe maglocks) and intrusion field-device load, at nominal current, per Section 28 00 00 (Addendum 6, TEECOM, May 14, 2025).

Role	Signature & Date	Notes
Prepared by (PM / Consultant)	___/___/___	Dynamic Security Technologies
Reviewed by _____	___/___/___	
GC Review (Devcon)	___/___/___	

— End of Report —